Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2021-P1-Model)** **Date: 04-02-2026**

Time: 09.00Am to 12.00Pm **GTA-14** **Max. Marks: 180**

KEY SHEET

PHYSICS

1	**A**	2	**C**	3	**C**	4	**B**	5	**2**	6	**2**
7	**1.73**	8	**5.69**	9	**81.6**	10	**96.60 – 96.80**	11	**BCD**	12	**BCD**
13	**ABD**	14	**BC**	15	**AD**	16	**BC**	17	**8**	18	**1**
19	**40**										

CHEMISTRY

20	**A**	21	**B**	22	**D**	23	**C**	24	**300**	25	**950**
26	**1950**	27	**500**	28	**4**	29	**6**	30	**ACD**	31	**ABC**
32	**ABCD**	33	**ABCD**	34	**ABD**	35	**CD**	36	**1**	37	**11**
38	**16**										

MATHEMATICS

39	**A**	40	**A**	41	**A**	42	**D**	43	**2.5**	44	**0.4**
45	**25**	46	**11**	47	**8**	48	**2**	49	**ABD**	50	**ABCD**
51	**ABCD**	52	**ABD**	53	**AB**	54	**ACD**	55	**20**	56	**7**
57	**1**										

For More Join: @IITJEE_Advanced

SOLUTIONS

PHYSICS

1. $P = P_0 + \frac{1}{2}\rho V^2 + \rho g(0.5)$

$P - P_0 = 10^3\left[\frac{30^2}{2} + 5\right] = 10^3[455] = 4.55 \times 10^5$

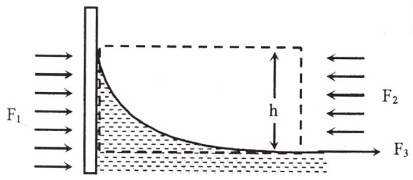
2. Let A cm^2 be the surface area of the bottom of the tank which is in contact with the furnace.
Volume of water evaporating in a time t (9 minute = 540 s)
= area x thickness of water layer = A x 1 = A cm^3
Mass of water evaporating in the time.
T = A x 1 = Ag
Amount of heat required to evaporate A g of water, i.e.,
Q = AL = A x 540 [as L, latent heat of steam = 540 cal/g]
= 540 A Cal .....(i)
Let T_1 be the temperature of the furnace and T_2 that of boiling water. Therefore,
$T_1 - T_2 = (T_1 - 100)$
Thickness of the tank, x = 2 cm
As $Q = \frac{kA(T_1 - T_2)t}{x}$ ..... (ii)
From eqns. (i) and (ii)
$\frac{kA(T_1 - T_2)t}{x} = 540A$ or $(T_1 - T_2) = \frac{540Ax}{kAt} = \frac{540x}{kt}$
or $T_1 - 100 = \frac{540 \times 2}{0.2 \times 540} = 10$ or $T_1 = 100 + 10 = 110^0 C$

3. $R = \frac{(300)^2}{100} = 900,\ \omega = 2\pi f = 2\pi \times \frac{150}{\pi} = 300$

$100 = \left[\frac{v^2}{R^2 + (\omega L)^2}\right]R \qquad 100 = \left[\frac{(500)^2}{(900)^2 + (300L)^2}\right]900$

$(300)^2 + (100)^2 L^2 = (500)^2 \qquad L^2 = 16,\ L = 4H$

4. The pressure of the water changes linearly with the increase in height. At the bottom of the meniscus it is equal to the external atmospheric pressure P_0, and at the top of $P_0 - gh$. The average pressure exerted on the wall is $P_{average}$ $p_0 - \square gh/2k$. The force corresponding to this value, for an aquarium with side walls of length l, is $F_1 = l p_{average} h$.



Consider the horizontal forces acting on the volume of water enclosed by the dashed lines in the figure. The wall pushes it to the right with force F_1, the external air pushed it to the left with force $F_2 = l p_0 h$, and the surface tension of the rest of the water pulls it to the right

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with a force $F_1=l$. The resultant of these forces has to be zero, since the volume itself is at rest. This means that $p_0-\left(p_0-\frac{1}{2}\rho gh\right)lh-p_0lh+l\gamma=0$

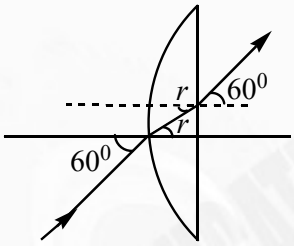
Which we can write as $h=\sqrt{\frac{2\gamma}{\rho g}}=\sqrt{\frac{2\times0.073}{1000\times10}}=0.0038\ m$

Water rises by approximated 4 mm up the wall of the aquarium.

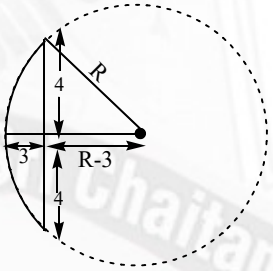
5. $V_0\propto\frac{Q}{a}\quad V\propto\frac{Q'}{2a},\frac{V_0}{V}=\frac{2Q}{Q'}=\frac{2\sigma a^2}{\sigma(2a)^2}=\frac{1}{2}$

6. The electric field at a point on the axis XX^1 which is very close to centre and above the plane is $\frac{Q}{n\in_0 a^2}=2$

7. & 8.



From the ray diagram it is clear that ray emerges out of lens parallel to itself. Hence the angle of deviation caused by the lens is zero. From snell's law we get $\sin 60°=\sqrt{3}\sin r\Rightarrow r=30°$. Since the emergent ray is parallel to initial incident ray, the portion of lens used for refraction can be assumed as slab. Hence lateral displacement $=\frac{t(\sin(i-r))}{\cos r}=\frac{3\sin(60°-30°)}{\cos 30°}=\sqrt{3}mm$. We have, $(R-3)^2+4^2=R^2\Rightarrow R=\frac{25}{6}$mm and $\frac{1}{f}=\frac{(\mu-1)}{R}\Rightarrow f=\frac{R}{(\mu-1)}=\frac{25}{6(\sqrt{3}-1)}=\frac{25}{12}(\sqrt{3}+1)mm$



9. Maximum energy of photon incident in H - sample = 13.6 eV + 27.2 = 40.8 eV this corresponds to the transition 2 to 1 in He^+ sample. For this electrons in He+ sample must be excited to a maximum of 1st excited state.
Since mass of -particle and He^+ ion is almost same, the -particles transfer 1/2 of their kinetic energy to excitation energy (40.8eV) for He^+ion.
excitation energy of He^+.=2 × 40.8 = 81.6 eV

10. The K.E. of particle should not be so large that it excites electron in He^+ to 2nd excited state. excitation energy of He^+. = 2 × 48.4 = 96.8 eV

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11. Restoring torque $\tau = -MB\sin\theta \left(\frac{mr^2}{2}\right)\alpha = -\left(\pi r^2 i_2\right)\left(\frac{\mu_0 i_1}{2R}\right)\theta \qquad \alpha = \frac{-\pi\mu_0 i_1 i_2}{mR}\theta$ SHM

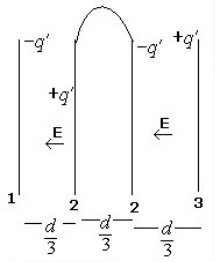
12. Given $E = \frac{\sigma}{\epsilon_0}$ $600 = \frac{q_0}{A\epsilon_0}$ (initially) $= \frac{\varepsilon}{d}$

Let final charge is q'.

$V_3 - V_2 = \frac{q'}{A\epsilon_0}\frac{d}{3}$ $V_3 - V_1 = \frac{2q'd}{3A\epsilon_0}$ $V_2 - V_1 = \frac{q'}{A\epsilon_0}\frac{d}{3}$ $q' = \frac{3}{2}\left(\frac{A\epsilon_0}{d}\right)\varepsilon$

But $q_0 = \frac{A\epsilon_0}{d}\varepsilon$ $q' > q_0$ capacitance increases at same p.d. ε.

Field between 1 & 2 $= \frac{q'}{A\epsilon_0} = \frac{3}{2}\frac{(A\epsilon_0)}{d}\frac{\varepsilon}{A\epsilon_0} = \frac{3}{2}\times 600 = 900\frac{V}{m}$



13. USE SYMMETRY

14. The error in the first student is less

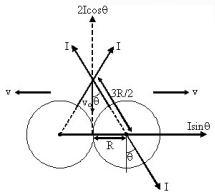
15. work done is directly proportional to $(gh + \mu gl)$

16. Power delivered to ship = Mau, Power delivered to water = $\frac{M}{3}au$

Total power = $\frac{4}{3}Mau + p_{viscous}$ $> \frac{4}{3}Mau$

18. $v = \sqrt{\frac{B}{\rho}}$

17. I= impulse by each of bigger balls on smaller ball sin θ =2/3, $\cos\theta = \frac{\sqrt{5}}{3}$ By impulse momentum theorem, for smaller ball 2 I $\cos\theta = Mv_0 \Rightarrow I\frac{3Mv_0}{2\sqrt{5}}$ Now, I sin θ=Mv (here v = speed acquired by each of the bigger balls after collision) applying law of conservation of energy, we get $\frac{1}{2}M\left(\frac{v_0}{\sqrt{5}}\right)^2 = \frac{1}{2}kx^2 \Rightarrow x = v_0\sqrt{\frac{M}{5K}} \Rightarrow x = 1.6\sqrt{\frac{1.25}{5\times 100}} \Rightarrow x = \frac{1.6}{20}m \Rightarrow x = 8cm$



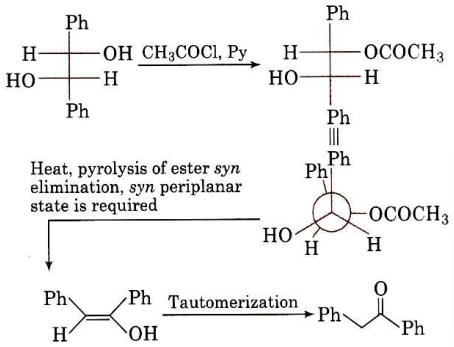
19. P.d across the central limits = P.d across 16Ω = P.d. across left limb = 16V
Currnet through 16Ω = 1A, Current through left limb = 1A $\therefore R = 11\Omega$

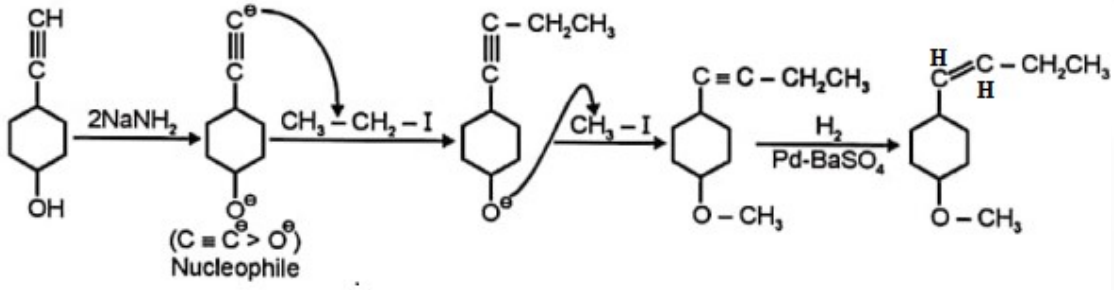
$$\tau_c = \left(\frac{16}{2} + 2\right)4\times 10^{-6} = 40\mu s$$

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CHEMISTRY

20.



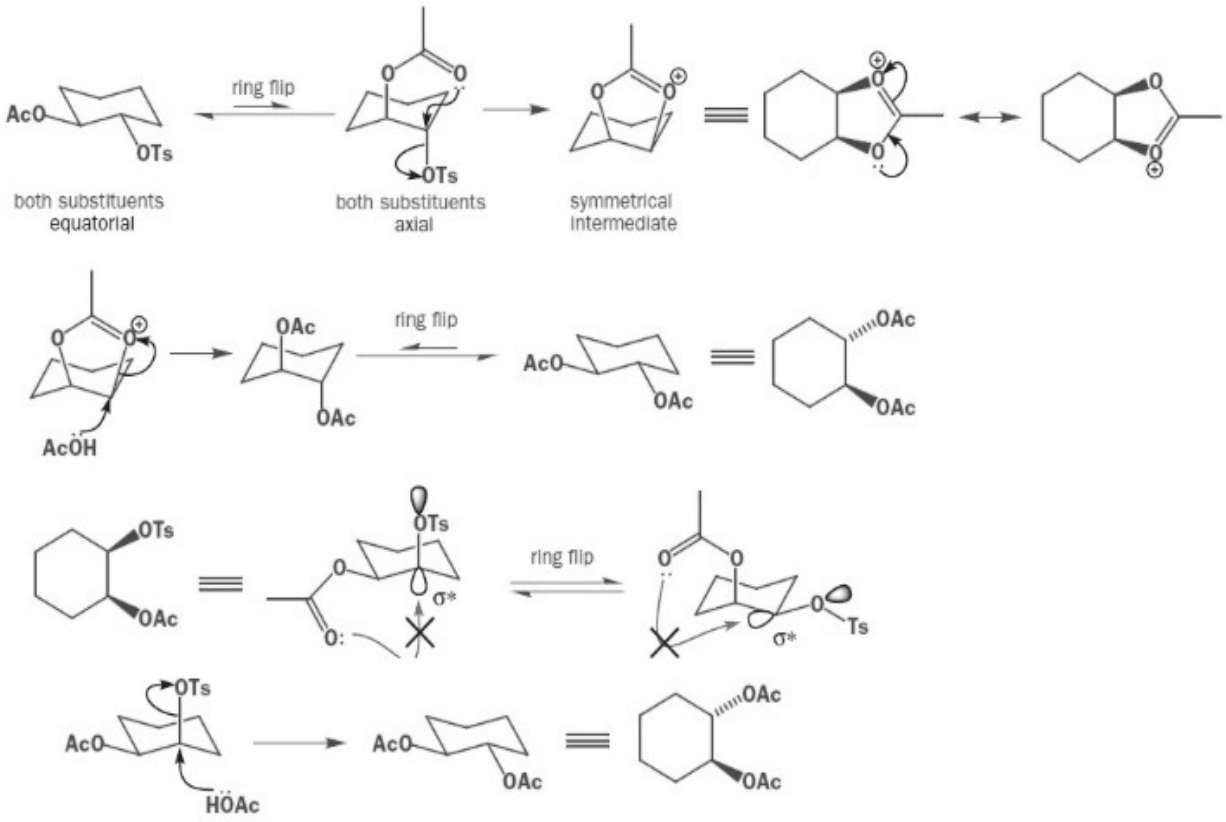


21.

22.

$$\Delta H_{reaction} = 4980 - 5328 = -348\ kJ / mol$$

23.



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24. & 25.

$$K=\frac{1}{t}.1n\frac{P_x^\circ}{P_x} \qquad \text{For } t=100 \text{ min}, K_1=\frac{1}{100}.1n\frac{800}{400}=\frac{1n2}{100}\text{min}^{-1}$$

For $t=200$ min, $K_2=\frac{1}{200}.1n\frac{800}{200}=\frac{1n2}{100}\text{min}^{-1}$

As $K_1=K_2$, order of reaction $=1$ $\qquad T_{87.5\%}=3t_{0.5}$

$$2X(g)\rightarrow 3Y(g)+2Z(g)$$

$t=0$	800	0	0	$=700 \therefore P_{total}=800+\frac{3}{2}x=950$ torr
$t=t$	$800-x$	$\frac{3}{2}x$	x	

26. & 27.

x Kg = Slurry
y Kg = Concentrated solution
2 Kg = water vapour
Total feed = 25,000 Kg
Step 2: NaCl (salt) balance
Given Feed contains 18% NaCl
Slurry contains 90% NaCl , NaCl goes only into slurry

NaCl in feed: $25000\times\frac{15}{100}=45000\ Kg\ NaCl$, NaCl in slurry: $x\times\frac{90}{100}$

Equate NaCl: $x\times\frac{90}{100}=4500\ \ x=\frac{4500\times100}{90}=5000\ Kg$

Slurry (x)=5000 Kg

Step 3: NaOH balance
Given: Feed contains 2% NaOH
Slurry contains 5% NaOH
Concentrated solution contains 50% NaOH

NaOH in feed: $25000\times\frac{2}{100}=500\ Kg\ NaOH$

NaOH in products: $x\times\frac{5}{100}+y\times\frac{5}{100}$

Substitute $x=5000$: $5000\times\frac{5}{100}+y\times\frac{50}{100}\ 250+0.5y$

Equate NaOH: $500=250+0.5y\ \ 0.5y=250\Rightarrow y=500\,Kg$

Concentrated solution $(y)=500\ Kg$

Step 4: Over all mass balance $x+y+z=25000$

$z=25000-(5000+500)\ z=19500\,Kg$, Water vapour $(z)=19{,}500\ Kg$

28. & 29.

number of radial nodes is equal to n-l-1

30. C_5H_8, degree of unsaturation = 2

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Reaction with Na in liq NH_3+ alkyl Iodide indicates a terminal alkyne (acetylide formation) P is $C_3H_7-C\equiv CH$

$$P \xrightarrow[H_2SO_4\ (dil)]{HgSO_4} \underset{(R)}{C_3H_7-\overset{\overset{O}{\|}}{C}-CH_3}$$

$$P \xrightarrow[liq\ NH_3]{Na} C_3H_7-C\equiv CNa \xrightarrow{C_3H_7I} C_3H_7-C\equiv C-C_3H_7 \xrightarrow{alk\ \ KMnO_4} 2C_4H_8O_2$$

31. In methoxy methane, The $C-O-C$ bond angle is slightly greater than the tetrahedral angle due to the repulsive interaction between the two bulky (–R) groups.

32. A) Keto carbon is reduced to chiral carbon. While aldehyde to achiral
B) Non-mirror images
C) C_4 epimers
D) They difference in configuration at C_3, C_4, C_5

33. The mineral $KCl.MgCl_2.6H_2O$ is know as carnallite. The iron obtained from Blast furnance is known as pig iron which contains about 4% carbon and many impurities (S, P, Si, Mn) in smaller amount. Froth floatation method of concentrating the ore is used for removing gangue from sulphide ores. Silver and gold are extracted from their ore by leaching process.

34. Distance between nearest octahedral and tetrahedral void:

Octahedral void at $\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$ and tetrahedral void at $\left(\frac{1}{4},\frac{1}{4},\frac{1}{4}\right)$

$$d = a\sqrt{\left(\frac{1}{4}\right)^2+\left(\frac{1}{4}\right)^2+\left(\frac{1}{4}\right)^2} = \frac{\sqrt{3a}}{4}$$

Distance between two nearest octahedral void is

Between the body-center octahedral void $\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$ and an edge-center void

$$\left(\frac{1}{2},\ 0,\ 0\right) d = a\sqrt{\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2} = \frac{a}{\sqrt{2}}$$

Nearest tetrahedral voids are separated by distance of a/2

In close packing interlayer spacing is $d = 2r\sqrt{\frac{2}{3}}$

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35. The Effective Atomic Number (EAN) is the total count of electrons surrounding the central metal atom in a coordination complex, including its own valence electrons and those donated by ligands, used to predict complex stability, often aiming for a noble gas configuration (like 36 or 54 electrons) it's calculated as: EAN = Atomic Number (Z) – Oxidation state + (2 x coordination Number), or simply counting the metal's electrons plus donated ligand electrons.

36. $Sb_2S_3(s) + 3H_2(g) \rightleftarrows 2Sb(s) + 3H_2S(g)$

Initial moles 0.01 0.01 0 0

Equ moles 0.01 – x 0.01 – 3x 2x $3x = \frac{1.19}{238} = 5\times10^{-3} = 5\times10^{-3}$

Now, $K_c = \frac{\left(5\times10^{-3}\right)^3}{\left(5\times10^{-3}\right)^3} = 1$

37. Initial m moles of $Cu^{2+} = 5$;

$m - eq.or\ m - moles\ of\ H^+$ produced

$100\times10^{-2} = 1 \Rightarrow m-$ moles of Cu^{2+} converted into $Cu = \frac{1}{2} = 0.5$

$m-$moles of Cu^{2+} remaining in solution $= 5 - 0.5 = 4.5$

$2Cu^{2+} + 4I^- \rightarrow Cu_2I_2 + I_2$ And $I_2 + 2NaS_2O_3 \rightarrow 2NaI + Na_2S_4O_6$

m moles of Cu^{2+} remaining $= m$ moles of $Na_2S_2O_3$

$4.5 = 0.04\times V \Rightarrow V = 112.5mL$

38. $\mathbf{n}CO_2 = 224 / 22400 = 0.01$ $\mathbf{n}H^+$ used $= (30 \times 1) / 1000 = 0.03$

$CO_2 + OH^- \rightarrow HCO_3^-$

0.01 mole (L.R.) 0.01 mole or less

Hence, moles of H^+ used should be 0.01 or less and titration of HCO_3^- and H^+ should not be detected by phenolphthalein. Hence, OH^- must be in excess.

$CO_2 + 2OH^- \rightarrow CO_3^{2-} + H_2O$

0.01 mole 0.02 mole 0.01 mole

Thus, 0.01 mole of CO_3^{2-} will require only 0.01 mole of H^+ in the presence of phenolphthalein.

As the mole of H^+ used is 0.03, 0.02 mole of OH^- must be present in excess.

Hence, total moles of OH^- used = 0.02 + 0.02 = 0.04

∴ [NaOH] used = 0.04 / 1 = 0.04 M

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MATHEMATICS

39. $\tan\left(\sin^{-1}\frac{3}{5}-2\cos^{-1}\frac{2}{\sqrt{5}}\right)=\tan\left(\tan^{-1}\frac{3}{4}-2\tan^{-1}\frac{1}{2}\right)$

$$\tan\left(\tan^{-1}\frac{3}{4}-\tan^{-1}\frac{2\times\frac{1}{2}}{1-\left(\frac{1}{2}\right)^2}\right)=\tan\left(\tan^{-1}\frac{3}{4}-\tan^{-1}\frac{4}{3}\right)$$

$$=\tan\left(\tan^{-1}\left(\frac{-7}{24}\right)\right)=\frac{-7}{24}$$

40. According to given data

$a_{11}=f\left(x^{2-1-2}\right)=f\left(\frac{1}{x}\right)$ | $b_{11}=f\left(x^{2-1-1}\right)=f\left(x^0\right)=f(1)$

$a_{21}=f\left(x^{4-1-2}\right)=f(x)$ | $b_{12}=f\left(x^{2-2-1}\right)=f\left(\frac{1}{x}\right)$

$a_{12}=f\left(x^{2-2-2}\right)=f\left(\frac{1}{x^2}\right)$ | $b_{21}=f\left(x^{4-1-1}\right)=f\left(x^2\right)$

$and\ a_{22}=f\left(x^{4-2-2}\right)=f(1)$ | $b_{22}=f\left(x^{4-2-2}\right)=f(x)$

$$\therefore A=\begin{bmatrix} f\left(\frac{1}{x}\right) & f\left(\frac{1}{x^2}\right) \\ f(x) & f(1)\end{bmatrix}\quad B=\begin{bmatrix} f(1) & f\left(\frac{1}{x}\right) \\ f\left(x^2\right) & f(x)\end{bmatrix}\Rightarrow B^T=\begin{bmatrix} f(1) & f\left(x^2\right) \\ f\left(\frac{1}{x}\right) & f(x)\end{bmatrix}$$

$$A+B^T=\begin{bmatrix} f\left(\frac{1}{x}\right)+f(1) & f\left(\frac{1}{x^2}\right)+f\left(x^2\right) \\ f(x)+f\left(\frac{1}{x}\right) & f(1)+f(x)\end{bmatrix}=\begin{bmatrix} -x^2f\left(x+f(1)\right) & \left(1-\frac{1}{x^4}\right)f\left(\frac{1}{x^2}\right) \\ f\left(\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right) & \frac{-1}{x^2}f\left(\frac{1+f(1)}{x}\right)\end{bmatrix}\text{(given)}$$

$$\Rightarrow f\left(\frac{1}{x}\right)+f(1)=-x^2f\left(x+f(1)\right)\quad\text{---------- (1)}$$

$$f\left(\frac{1}{x^2}\right)+f\left(x^2\right)=\left(1-\frac{1}{x^4}\right)f\left(\frac{1}{x^2}\right)\quad\text{---------- (2)}$$

$$f(x)+f\left(\frac{1}{x}\right)=f\left(\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right)\quad\text{---------- (3)}$$

$$f(1)+f(x)=\frac{-1}{x^2}f\left(\frac{1+f(1)}{x}\right)\quad\text{---------- (4)}$$

Putting $x=1$, in (2), we get

$2f(1)=0\Rightarrow f(1)=0$ ------- (5)

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From (1): $f\left(\frac{1}{x}\right)=-x^2 f(x)$ ---*

So now $\int_x^{\frac{1}{x}} f(t)dt = I$ (Let) --- (1)

Put $t=\frac{1}{z}$ | Lower Limit at $t=x$, Then $z=\frac{1}{x}$

$dt=-\frac{1}{z^2}dz$ | Upper Limit at $t=\frac{1}{x}$, Then $z=x$

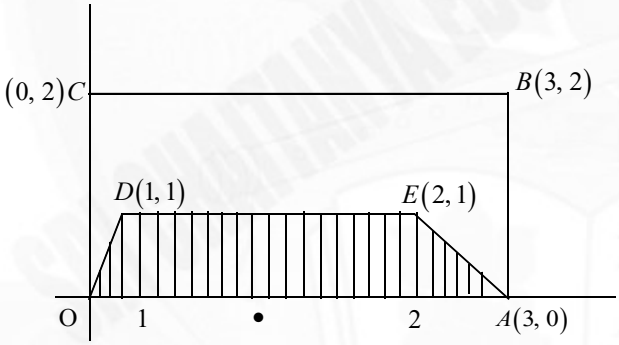
$$I=\int_{\frac{1}{x}}^{x} f\left(\frac{1}{z}\right)\left(\frac{-1}{z^2}dz\right) \quad =\int_x^{\frac{1}{x}} f\left(\frac{1}{z}\right)\left(\frac{1}{z^2}\right)dz$$

$$\Rightarrow I=\int_x^{\frac{1}{x}} f\left(\frac{1}{t}\right)\frac{1}{t^2}\,dt \quad \text{----- (2)}$$

$$2I=\int_x^{\frac{1}{x}}\left(f(t)+\frac{1}{t^2}f\left(\frac{1}{t}\right)\right)dt \quad =\int_x^{\frac{1}{x}}\left(f(t)+\frac{1}{t^2}\left(-t^2 f(t)\right)\right)dt \quad \text{from *}$$

$2I=0 \qquad \Rightarrow I=0$

41.



i) $0\le x\le 1$ and $0\le y\le 1 \quad \therefore d(P, OA)\le d(P, OC)\Rightarrow y\le x \qquad 1\le x\le 2$

ii) $d(P, OA)\le d(P, BC) \qquad \Rightarrow y\le 2-y\Rightarrow y\le 1$

42. (d): Let $\log_{10} x=a,\ \log_{10} y=b,\ \log_{10} z=c$

$a+b-ab=4-\log_{10} 2000 = \log_{10} 5$

$b+c-bc=1-\log_{10} 2 = \log_{10} 5$

$c+a-ca = 0$

(i), (ii) $\Rightarrow (c-a)(b-1) = 0 \Rightarrow c = a$ and (iii) $\Rightarrow 2c=c^2\Rightarrow c=0, 2$

$a=c=0\Rightarrow x_1=1,\ z_1=1,\ y_1=5$ & $a=c=2$

$\Rightarrow x_2=100,\ z_2=100,\ y_2=20 \qquad \therefore y_1+y_2=5+20=25$

43 & 44.

$f(x)$ is symmetric about $x=\frac{7}{4} \qquad \frac{-b}{2a}=\frac{7}{4}\Rightarrow\frac{-b}{2a}=\frac{7}{2}$

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$ax^2+bx+a=7x+a$ has only one real solution

$ax^2+x(b-7)=0$ has discriminant = 0

$(b-7)^2-4a(0)=0 \qquad b-7 \Rightarrow a=-2 \qquad \frac{a+b}{2}=\frac{5}{2}=2.5$

$f_{\min}=f(0)=a=-2 \quad \left|\frac{2}{5}\right|=0.4$

45. Let the slope of line Ap be m

Equation of AP is $y=mx+8$

Equation of BP in $y=-\frac{x}{m}-2$

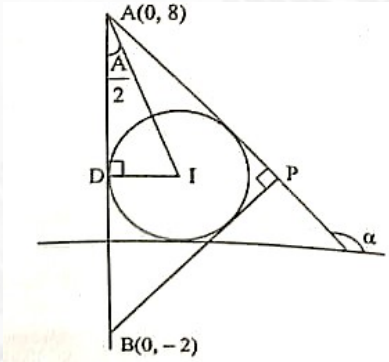
Solving both equations

$P\equiv\left(\frac{-10m}{m^2+1},\frac{-2m^2+8}{m^2+1}\right)$

For maximum abscissa of P

$x\equiv\frac{-10}{m+\frac{1}{m}}=\frac{-10}{-2}=5(m=-1) \;\therefore$ Area of $\Delta APB=\frac{1}{2}\times10\times5=25\ sq\ units$

46. Incentre $\left(\frac{5\sqrt{3}-5}{2},\frac{1+5\sqrt{3}}{2}\right)$



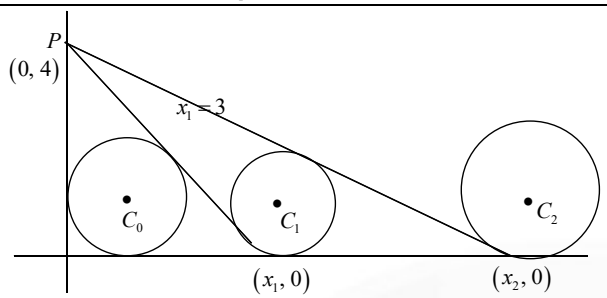
$\tan\frac{A}{2}=\frac{ID}{AD}=\frac{\frac{5\sqrt{3}-5}{2}}{8-\left(\frac{1+5\sqrt{3}}{2}\right)}=\frac{5(\sqrt{3}-1)}{15-5\sqrt{3}}=\frac{1}{\sqrt{3}} \quad \frac{A}{2}=30^0\Rightarrow A=60^0$

Inclination of line AP is $a=150^0 \qquad \therefore$ Slope of line AP is $\tan(150^0)=\frac{-1}{\sqrt{3}}$

Equation of AP is $y-8=\frac{-x}{\sqrt{3}} \quad \Rightarrow x+\sqrt{3}y=8\sqrt{3}$

47 & 48.

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Let $C_n = (C_n, 1)$ be the centres of the circles and $X_n = (x_n, 0)$ the intersection of Tangent lines with x-axis (see fig) As $C_{n-1} X_n C_n$ is a right angle Δ^{le} $(bcz\ \ C_{n-1}X_n\ \ \&\ \ X_nC_n\ \ are\ Bisectors\ of\ \ Adjacent\ \ angles)$ Then by the Geometric Mean theorem the square of its altitude from X_n (which has unit length) equals the product of the '2' parts into which the altitude divided the hypotenuse i.e., $(x_n - C_{n-1})(c_n - x_n) = 1$

If we set $\delta_n = C_n - x_n$, above is reduced to $x_n - C_{n-1} = \frac{1}{\delta_n}$

and $C_n - C_{n-1} = \frac{1}{\delta_n} + \delta_n$ --- (1)

Other hand of T_n is the Tangency point in fig below, we have

$PT_n = Px_n - \delta_n$ & $PT_n^2 = PC_n^2 - 1$

$\Rightarrow \left(\sqrt{(C_n - \delta_n)^2 + 16} - \delta_n\right)^2 = C_n^2 + 9 - 1 \Rightarrow C_n = \frac{2}{\delta_n} - \delta_n$ --- (2)

Putting this into (1) we get

$C_{n-1} = \frac{1}{\delta_n} - 2\delta_n$ & $\frac{1}{\delta_{n+1}} - 2\delta_{n+1} = \frac{2}{\delta_n} - \delta_n$ i.e., $\delta_{n+1} = \frac{1}{2}\delta_n$

$\delta_1 = \frac{1}{2}$ it follows that $\delta_n = \frac{1}{2^n}$

$x_n = C_n - \delta_n = \frac{2}{\delta_n} - 2\delta_n = 2^{n+1} - \frac{1}{2^{n-1}}$

Hence $\underset{n\to\infty}{Lt}\ \frac{x_n}{2^n} = 2$

Centre $(C_2, 1)$

Then (2) $C_n = \frac{2}{\delta_n} - \delta_n$

$C_2 = \frac{2}{\delta_2} - \delta_2 = \frac{2}{1/2^2} - \frac{1}{2^2} = 8 - \frac{1}{4} = \frac{31}{4}$

$C_2\left(\frac{31}{4}, 1\right)$ $\therefore p + q = \frac{31}{4} + 1 = \frac{35}{4} \cong 8.75$

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This is a known configuration when ΔPOX_n with height $h=|OP|$ and in radius 'r' is split into 'n' *triangle's* .

$\Delta Pox_1, \Delta POX_1X_2, ---, \Delta PX_{n-1}X_n,$ such that all their in circles have the same radius r_n

The radius 'r' of the inscribed circles of ΔPox_n is found as

$$r=\frac{h}{2}\left(1-\left(1-\frac{2rn}{h}\right)^n\right) \quad \text{------ (1)}$$

For $h=4, r_n=1,$ we have $r=2-2^{1-n}$ ---- (2)

$$2r=|OP|+|OX_n|-|PX_n| \quad =h+x_n-\sqrt{h^2+x_n^2}$$

$$x_n=\frac{2r(h-r)}{h-2r}=2^{n+1}-2^{1-n}$$

$$\underset{n\to\infty}{Lt}\frac{x_n}{2^n}=\frac{2^{n+1}-2^{1-n}}{2^n}=2$$

49. A) Let e_1 and e_2 be the eccentricities of ellipse and hyperbola, then

$$a^2=\lambda^2a^2\left(1-e_1^2\right) \text{ Or } e_1=\sqrt{\left(1-\frac{1}{\lambda^2}\right)} \text{ and } e_2=\sqrt{2}$$

Now, $\lambda ae_1=ae_2 \quad \Rightarrow \lambda\sqrt{\left(1-\frac{1}{\lambda^2}\right)}=\sqrt{2}\Rightarrow\lambda^2-1=2 \text{ or } \lambda=\pm\sqrt{3}$

$$\therefore \lambda=\sqrt{3} \quad \frac{e_1}{e_2}=\frac{1}{\lambda}=\frac{1}{\sqrt{3}}$$

B) Any tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ forms a triangle with the asymptotes which has constant area ab.

Given $ab=a^2 \tan\lambda$ or $b=a\tan\lambda$

or $b^2=a^2\tan^2\lambda$ or $a^2\left(e^2-1\right)=a^2\tan^2\lambda$

or $e^2\equiv 1+\tan^2\lambda\equiv\sec^2\lambda\ \therefore\ e^2\equiv\sec\lambda$

C) $x^2+y^2=a^2$ and $xy=c^2$,

We have, $x^2+\frac{c^4}{x^2}=a^2$ or $x^4-a^2x^2+c^4=0$ or $\sum x_i=0$ *and* $\prod x_i=c^4$

Similarly, if we eliminate x, then $Y^4-a^2y^2+c^4=0$ or $\sum y_i=0$ *and* $\prod x_i=c^4$

D) Since $(a\sec\theta, b\tan\theta)$ lies on

$$y=mx+\sqrt{\left(a^2m^2-b^2\right)} \therefore b\tan\theta=am\sec\theta+\sqrt{\left(a^2m^2-b^2\right)}$$

$$\Rightarrow\left(b\tan\theta-am\sec\theta\right)^2=a^2m^2-b^2$$

$$\Rightarrow a^2m^2\tan^2\theta-\frac{2abm\sin\theta}{\cos^2\theta}+b^2\sec^2\theta=0$$

or $a^2m^2\sin^2\theta-2abm\sin\theta+b^2=0 \qquad (\because \cos\theta\neq 0)$

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$$\therefore \sin\theta = \frac{2abm \pm \sqrt{4a^2b^2m^2 - 4a^2b^2m^2}}{2a^2m^2} = \left(\frac{b}{am}\right) \therefore \theta = \sin^{-1}\left(\frac{b}{am}\right)$$

50. A flea starts from 0.

- 1st jump: $+\frac{1}{2}$ (right)
- 2nd jump: $+\frac{1}{4}$ (left)
- 3rd jump: $+\frac{1}{8}$ (right)
- 4th jump: $+\frac{1}{16}$ (left)
- and so on ….

So the jumps form an alternating geometric series.

Let position after n jumps be t_n.

Step 1: Write general terms

Magnitude of jumps:

$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \ldots\ldots$

Direction alternates:

$t_n = \frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} + \ldots.(\text{n terms})$

This a GP with:

- First term $a = \frac{1}{2}$
- Common ration $r = -\frac{1}{2}$

Step 2: Formula for sum of GP

$$t_n = S_n = \frac{a\left(1 - r^n\right)}{1 - r}$$

Substitute values:

$$t_n = \frac{\frac{1}{2}\left(1 - \left(-\frac{1}{2}\right)\right)^n}{1 + \frac{1}{2}}$$

$$t_n = \frac{\frac{1}{2}\left(1 - \left(-\frac{1}{2}\right)^n\right)}{\frac{3}{2}}$$

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$$t_n = \frac{1}{3}\left(1-\left(-\frac{1}{2}\right)^n\right)$$

Step 3: Verify given options

Option $A: t_9 = \frac{171}{512}$

$$t_9 = \frac{1}{3}\left(1+\frac{1}{512}\right) = \frac{1}{3}.\frac{513}{512} = \frac{171}{512}$$

Option $B: t_{10} = \frac{341}{1024}$

$$t_{10} = \frac{1}{3}\left(1-\frac{1}{1024}\right) = \frac{1}{3}.\frac{1023}{1024} = \frac{341}{1024}$$

Step 4: Sum of position

We are given expression for

$\sum_{r=1}^{n} t_r$

Using GP sum results:

When n is odd:

$$\boxed{\sum_{r=1}^{n} t_r = \frac{n}{3}+\frac{1}{9}\left(\frac{2^n+1}{2^n}\right)}$$

When n is even:

$$\boxed{\sum_{r=1}^{n} t_r = \frac{n}{3}+\frac{1}{9}\left(\frac{2^n-1}{2^n}\right)}$$

So,

Option C (n odd) correct

Option D (n even) correct

Final Answer

All option A, B, C and D are correct

51. A) Required probability

$$= \frac{5!\left(\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}\right)}{5_{P_5}} = \frac{44}{120} = \frac{11}{30}$$

A) Required probability

$$= \frac{2\times {}^5C_4}{5^4} = \frac{2}{125}$$

$7 \to (1,1,1,1,3)$ (or) $(1,1,1,2,2)$

C) No. of onto functions

$$= \frac{7!}{3!}\times\frac{5!}{4!}+\frac{7!}{2!2!}\times\frac{5!}{3!\times 2!} = \frac{7!\times 5!}{36}$$

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Required probability $= \dfrac{7! \times 5!}{\dfrac{36}{5^7}} = \dfrac{7! \times 2}{5^6 \times 3}$

D) Case (i)

$f(0) = f(1) = f(2)$

No. of functions $= {}^8C_1$

Case (ii)

$f(0) = f(1) < f(2)$

NO. of functions $= {}^8C_2$

Case (iii)

$f(0) < f(1) < f(2)$

NO. of functions $= {}^8C_3$

Required probability

$= \dfrac{{}^8C_1 + {}^8C_2 + {}^8C_2 + {}^8C_3}{8^3}$

$= \dfrac{{}^{10}C_3}{8^3} = \dfrac{15}{64}$

52. $n_1 = 10.10.10 = 1000$, as each of i, j *and* k have 10 choices.

For S_2, we do casework on j.

Given, $i < j+2$ i.e., $i \le j+1$

$j = 1$ $\quad i = 1,2$ $\quad \rightarrow 2$ *choices*

$j = 2$ $\quad i = 1,2,3$ $\quad \rightarrow 3$ *choices*

....

$j = 8$ $\quad i = 1,2,3,...,9$ $\quad \rightarrow 9$ *choices*

Hence, $n_2 = 2+3+..+9 = (1+2+...+9)-1$

$= \dfrac{9 \times 10}{2} - 1 = 45 - 1 = 44$

$n_3 = {}^{10}C_4 = \dfrac{10.9.8.7}{24} = 210$

$n_4 = {}^{10}C_4 . = 4! = 210.4!$

Hence, $\dfrac{n_4}{12} = 420$

53. $T_n = \dfrac{1}{\log\left(\prod_{r=1}^{n} \dfrac{r^2}{1+r^2}\right)}$

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$$\therefore \prod_{r=1}^{n} \frac{r^2}{1+r^2} = \left(\frac{(n!)^2}{\prod_{r=1}^{n} 1+r^2} \right) \qquad \left(\because \prod_{r=1}^{n} r^2 = (1.\ 2.\ 3....n)^2 \right)$$

$$\log\left(\prod_{r=1}^{n} \frac{r^2}{1+r^2} \right) = 2\log_e(n!) - \sum_{r=1}^{n} \log\left(1+r^2\right)$$

Now we know that $\boxed{\frac{x}{1+x} \le \log(1+x) \le x}^{*}$

Consider $\log(1+x) \le x$ clearly $D_n = 2\log_e(n!) - \sum_{r=1}^{n} \log\left(1+r^2\right)$

$$\Rightarrow \log\left(1+r^2\right) \le r^2$$

$$\Rightarrow -\log\left(1+r^2\right) \ge -r^2$$

$$\Rightarrow -\sum_{r=1}^{n} \log\left(1+r^2\right) \ge \sum_{r=1}^{n} -r^2$$

Adding $2\log(n!)$ on both sides we get

$$2\log(n!) - \sum_{r=1}^{n} \log\left(1+r^2\right) \ge 2\log(n!) - \sum_{r=1}^{n} r^2$$

$D_n \ge 2\ln(n!) - \sum_{r=1}^{n} r^2$ lower bond ------ (1)

Now $\log(1+x) \ge \frac{x}{1+x}$

$$-\log(1+x) \le -\frac{x}{1+x}$$

$$-\log\left(1+r^2\right) \le -\frac{r^2}{1+r^2}$$

$$-\sum_{r=1}^{n} \log\left(1+r^2\right) \le -\sum_{r=1}^{n} \frac{r^2}{1+r^2}$$

Add $2\log(n!)$ on both sides, we get

$$2\log(n!) - \sum_{r=1}^{n} \log\left(1+r^2\right) \le 2\log(n!) - \sum_{r=1}^{n} \frac{r^2}{1+r^2}$$

$D_n \le 2\log(n!) - \sum_{r=1}^{n} \frac{r^2}{1+r^2}$ upper bond ----- (2)

So from above two inequalities

$$2\log(n!) - \sum_{r=1}^{n} r^2 \le D_n \le 2\log(n!) - \sum_{r=1}^{n} \frac{r^2}{1+r^2}$$

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$$\Rightarrow \frac{1}{2\log(n!)-\sum_{r=1}^{r}\frac{r^2}{1+r^2}} \le T_n \le \frac{1}{2\log(n!)-\sum_{r=1}^{r} r^2}$$

A) According to the option $\frac{4n}{(n-1)^2-n^2\log_e(4)} = \frac{4n}{(n-1)^2-n^2(1.38)}$

Since $\log_e 4 \cong 1.38$

So denominator $n^2+1-2n-1.38n^2 \cong 1-2n$

$\therefore \underset{n\to\infty}{Lt}\frac{4n}{1-2n} \cong -2._$

$\le$ *small* $(-ve)(-0._)$

as $n \to \infty$, D_n becomes more $(-Ve)$

$\left(\sum r^2 >>> \log(n!)\right)$

So $T_n > RHS$

(or)

$D_n = 2\log n! - \sum\log(1+r^2)$

$D_n = \frac{1}{T_n}$ ------- (1)

$(n-1)^2 - n^2\log 4 > 4n\left(2\log(n!)-\sum_{r=1}^{n}\log(1+r^2)\right)$

$\log(1+r^2) \ge \log r^2$

(or)

$\log(1+r^2) \ge 2\log r$

$\sum_{r=1}^{n}\log(1+r^2) \ge \sum_{r=1}^{n} 2\log r$

$\Rightarrow D_n = 2\log(n!)-\sum_{r=1}^{n}\log(1+r^2)$

$\boxed{D_n \le 0}$

RHS: non-positive $(ie\ 0\ \&-ve)$ but it grows as n^2 LHS is $-ve$

(B)

$H_{100} = \sum_{k=1}^{100}\frac{1}{K} = H_{100}$ (Since Harmonic number)

$H_n = 1+\frac{1}{2}+\frac{1}{3}+-----\infty$

$S = \log(H_{100})$

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$f(x)=\frac{1}{x}$

If $K \in Z^+$

$\frac{1}{k+1}<\frac{1}{x}<\frac{1}{k} \quad \forall\ x \in [k, k+1]$

$\int_k^{k+1} \frac{dx}{k+1}<\int_k^{k+1} \frac{1}{x}dx<\int_k^{k+1} \frac{1}{k}dx$

$\frac{1}{k+1}<\log(k+1)-\log(k)<\frac{1}{k}$

$\sum_{k=1}^{n} \frac{1}{k+1}<\sum_{k=1}^{n} \left(\log_e(k+1)-\log_e k\right)<\sum_{k=1}^{n} \frac{1}{k}$

$k=1.\ \ H_n-1+\frac{1}{n+1}\ (Negligible)<\log_e(n+1)<H_n$

$\Rightarrow \cong\ H_n-1<\log_e(n+1)<H_n$

$H_n<1+\log_e(n+1)$

$H_n>\log_e(n+1)$

For higher bounds

$\log_e(1+n)\cong\log_e n, \qquad$ and $\qquad H_n<1+\log_e(n)$

$\therefore \log_e(n+1)<H_n<1+\log_e n$

$n=100, \qquad \log_e(101)<H_{100}<1+\log_e 100$

log on both $\qquad \log_e(\log_e(101))<S<\log_e(1+\log_e 100)$

$\log_e 100 \cong 2\log_e 10 \cong 2\times 2.302 \cong 4.604 \Rightarrow 1+\log_e 100 \cong 5.604$

And $\log_e(101)\cong 4.61 \quad \Rightarrow \quad \log_e(\log_e 101)\cong\log_e(4.61)$

$\log_e(1+\log_e 100)\cong\log_e(5.604)$

$\Rightarrow \boxed{\log_e(4.61)<S<\log_e(5.604)}$ $\left(\because 4.61<5.604 \Rightarrow \log_e(4.61)<\log_e(5.604)\right)$

So RHS: $\frac{100}{99}\log_e 50 = (1.01)(\log_e 5+\log_e 10)=1.01[1.-+2.302]$

$\cong 1.01(3.-)$

$\cong 3.-$

$\log_e(4.61)\cong 1.4<S<1.7$

$\downarrow = 1.6$

Which mean (B) $S<\frac{100}{99}\log_e 50$ is correct

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C) $T_n < \dfrac{2}{n(1-\log_e 2)-\cos^{-1}\left(\dfrac{2n}{1+n^2}\right)}$

$\cos^{-1}\left(\dfrac{2n}{1+n^2}\right)=2\,\mathrm{Tan}^{-1}\left(\dfrac{1-n}{1+n}\right)$ *when* $n\geq 1$

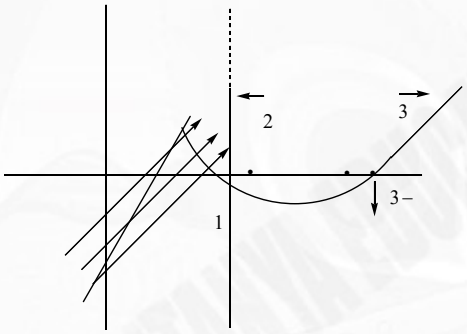
$D_n>0,\ n(1-\log_e 2)>0$ *for* $n=1$ Applicable for only n = 1

Don't go through putting value n directly

At $n\geq 2$ $\cos^{-1}\left(\dfrac{2n}{1+n^2}\right)$ grows faster then $n(1-\log_e 2)$

$g(x)=x(1-\log_e 2)-\cos^{-1}\left(\dfrac{2x}{1+x^2}\right)$

$g^1=(1-\log_e 2)-\left(\dfrac{1+x^2}{1-x^2}\right)=0$



D_n is increasing $\dfrac{2}{D_n} >> more\,(-Ve)$

$2\log n!-\sum_{r=1}^{n} r^2<0,\quad n=2 \quad \cong -4.- \quad$ for n = 2

$RHS=\dfrac{2}{2(1-\log_e 2)-\cos^{-1}\left(\dfrac{4}{5}\right)} \quad \cong -63.7$

$=-4._{-}<-63.7 \quad False \quad T_n>-$

D) $S+T_n>\dfrac{4n+100}{\dfrac{99}{100}+n\log 4}$

$S+T_n\cong 1.6+(-Ve)<2$

T_n is $-Ve$ & increase towards the +ve side as 'n' grows.

$RHS=\dfrac{4n+100}{\dfrac{99}{100}+n\log 4}\cong\dfrac{4n+100}{0.101+n(2.376)}\cong\dfrac{4}{2.376}$

$n+\infty^{+} \quad \cong 1.68=1.6+(-Ve)<1.68$ so D is wrong

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54. $f(x)=\int_0^x e^{x-y}f'(y)dy-\left(x^2-x+1\right)e^x,$

Let $I(x)=\int_0^x e^{x-y}f'(y)dy \Rightarrow I'(x)=e^x\int_0^x e^{-y}f'(y)dy+f'(x)$

$\Rightarrow I'(x)=\int_0^x e^{x-y}f'(y)dy+f'(x)$

$\therefore f(x)=I(x)-\left(x^2-x+1\right)e^x, \qquad \Rightarrow I'(x)=I(x)+f'(x)$

$\therefore\ I(x)'=f(x)+\left(x^2-x+1\right)e^x \Rightarrow I'(x)-I(x)=f'(x)$

$\Rightarrow\left[f(x)+\left(x^2-x+1\right)e^x\right]-\left[f(x)+\left(x^2-x+1\right)\right]$

$\Rightarrow f'(x)+e^x(2x-1)+\left(x^2-x+1\right)e^x-f(x)-\left(x^2-x+1\right)e^x=f'(x)$

$\Rightarrow \boxed{f(x)=e^x(2x-1)}$

Options: A) $f'(x)=e^x(2)+(2x-1)e^x=e^x(2x+1)$

$f''(x)=e^x(2x+3) \quad f'(x)=0\Rightarrow x=-\frac{1}{2}$

$f''(x)=e^{-\frac{1}{2}}(2)\geq 0 \qquad \therefore$ Local minimum at $x=-\frac{1}{2}$

$\therefore\ f\left(-\frac{1}{2}\right)=e^{-\frac{1}{2}}\left(2\left(-\frac{1}{2}\right)-1\right)=-2e^{-\frac{1}{2}}$

B) $f(x)>1-x+\frac{3x^2}{2}-\frac{5x^3}{6}$ as $x\to 0+$ $LHS\ of\ f(x)\cong -1$

$RHS\ of\ f(x)\cong +1$ So 'B' is along

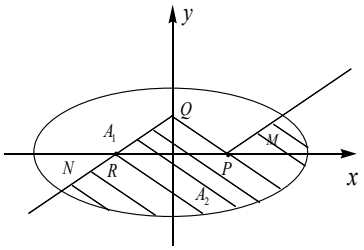
C) $\int_0^1 f(1-x)\,dx=\int_0^1 f(t)\,dt \quad \begin{matrix} t=1-x \\ dt=-dx \end{matrix} \Big| \begin{matrix} L.L & x=0, & t=1 \\ U.L & x=1, & t=0 \end{matrix}$

D) $f'(\beta)=2e^\beta \Rightarrow (2\beta+1)e^\beta=2e^\beta \Rightarrow 2\beta+1-2=0\ \beta=\frac{1}{2}$

So 'D' will be True provided that $\frac{1}{2}\in(0,x) \qquad \Rightarrow x>0, \quad x>\frac{1}{2}$

55. Area of an Ellipse $=\pi ab=\frac{20}{3}$ Area $\Delta le\,PQR=1$

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$A_1 = \frac{7}{3}\ A_2 = \frac{13}{3}$ (By Symmetric along M = area of N)

56. $a = 1000 = 2^3 \times 5^3$

$b = 2000 = 2^4 \times 5^3$

$\because$ L.C.M of (a, b) = 1000

They at the most contains 2 three times and 5 three times

Also the L.C.M of terms containing c is 2000 the term 'c' should contain 2 four times, as a and b can contain 2 only 3 times.

So 'c' should be $2^4 = 16,\ 2^4 \times 5^1 = 80,\ 2^4 \times 5^2 = 400,\ 2^4 \times 5^3 = 2000$

Either 'a' or 'b' should have 2^3.

Hence fixing one as 2^3 other can take values from 2^0 *to* 2^3

Hence total combinations $= 2 \times 4 - 1 = 7$ (the one combination is $2^3, 2^3$ which we took twice) 2 of the three should have power of 5 *as* 5^3.

Whereas the third can take 4 values in terms of powers of $5\left(5^0 \text{ to } 5^3\right)$

$\therefore$ Hence total no of such arrangement $\frac{3!}{2!} \times 4 - 2 = 12 - 2 = 10$ (this is for $5^3, 5^3, 5^3$ which we took thrice)

$\therefore$ Ordered triplets 7 x 10 = 70

57. $\frac{x^{2023} + 2024x + 2025}{x^2 - x + 3}\left(\frac{\sin x + 2}{e^{\pi x}}\right) = 0$

$\left(\sin x + 2, e^{\pi x}, x^2 - x + 3 \text{ Can never be equal to zero, always + ve}\right)$

$x^{2023} + 2024x + 2025 = 0$

Let $g(x) = x^{2023} + 2024x + 2025$

$g'(x) = (2023)x^{2022} + 2024 > 0, \forall x \in \mathbb{R}$

$g(x)$ Is an increasing function

$\therefore f(x) = 0$ Has only one solution.

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